

A Bernoulli Wreath Product Subcase of the Crann–Tanko QSIN Conjecture

fa_banach_001, agent_lane_03

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Status

This is a partial result packet for arXiv:1602.05259, Jason Crann and Zsolt Tanko, *On the operator homology of the Fourier algebra and its cb-multiplier completion*. The source conjectures that relative operator 1-biflatness of $A(G)$ is equivalent to G being QSIN. The theorem below proves the conjecture for compact Bernoulli wreath products $K_0^\Gamma \rtimes \Gamma$.

Source Problem

The source paper proves that $A(G)$ is relatively 1-biflat if and only if the diagonal G_Δ has a contractive approximate indicator in $B(G \times G)$, and then asks whether this is equivalent to QSIN. The relevant source crop is:

weakly for $v \in I(G_\Delta)$. Passing to convex combinations, and noting that $(\varphi_\gamma) \subseteq \mathcal{P}_1(G \times G)$, we obtain a contractive approximate indicator for G_Δ in $\mathcal{P}_1(G \times G)$. $\square$

We conjecture that $A(G)$ is relatively 1-biflat if and only if G is QSIN, meaning $L^1(G)$

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has a bounded approximate identity (f_α) satisfying

$$\|\beta_1(s)f_\alpha - f_\alpha\|_{L^1(G)} \rightarrow 0, \quad s \in G.$$

By Theorem [3.4] and [25, Corollary 3.2], for any locally compact group G such that $VN(G)$ is 1-injective in $\mathbb{C}\text{-mod}$, relative 1-biflatness of $A(G)$ implies that G is amenable, and therefore QSIN by [27, Theorem 3]. Hence, the conjecture is valid for all G such that $VN(G)$ is an injective von Neumann algebra, in particular, for any type I or almost connected group (cf. [31]). We now establish the conjecture for totally disconnected groups.

The source also proves the following semidirect-product obstruction, which is the main input used here:

and $\text{supp}(\omega_\alpha) \rightarrow \{e\}$. Hence, G is co-null. $\square$

For the semidirect product of an infinite compact group K by a discrete group H , we now show that relative 1-biflatness of $A(K \rtimes H)$ entails that the unitary representation

$$\pi_K : H \rightarrow \mathcal{B}(L_0^2(K)) : h \mapsto [\xi \mapsto h \cdot \xi]$$

weakly contains the trivial representation. Here, $L_0^2(K) = \{\xi \in L^2(K) : \int_K \xi = 0\}$ and $h \cdot \xi(k) = \xi(h^{-1}kh)$ for $h \in H$ and $\xi \in L^2(K)$, where $h^{-1}kh$ is the product in $K \rtimes H$, i.e. the action of h^{-1} on k . If, moreover, the action of H on K is ergodic, we show that the Haar integral on K is not the unique H -invariant mean on $L^\infty(K)$. *Ergodicity* of the H -action on K is the assertion that if $E \subseteq K$ is Borel with $E \triangle h \cdot E$ null for all $h \in H$, then E must be null or co-null, and is equivalent to the non-existence of normal H -invariant means on $L^\infty(K)$ other than 1_K .

Proposition 4.3. *Let $K \rtimes H$ be the semidirect product of an infinite compact group K by a discrete group H . If $A(K \rtimes H)$ is relatively 1-biflat, then π_K weakly contains the trivial representation.*

PROOF. Let G denote $K \rtimes H$. As in the proof of Theorem 4.1, relative 1-biflatness of $A(G)$ yields a net of states (ω_{ξ_α}) in $A(G)$ with $\xi_\alpha \in \mathcal{P}_G$ satisfying

$$\|v \cdot \omega_{\xi_\alpha} - v(e)\omega_{\xi_\alpha}\|_{A(G)}, \quad \|s \triangleleft \omega_{\xi_\alpha} - \omega_{\xi_\alpha}\|_{A(G)} \rightarrow 0, \quad v \in A(G), \quad s \in G.$$

Arguing as in the proof of Proposition 4.2 we may assume $\text{supp}(\omega_{\xi_\alpha}) \rightarrow \{e\}$ and, since K is an open subgroup of G , we may identify $A(K)$ with a subspace of $A(G)$ and further assume that $\text{supp}(\xi_\alpha) \subseteq K$. Viewing $L^2(K)$ as a subspace of $L^2(G)$ via extension by zero, we have $\beta_2^G(h)\xi = h \cdot \xi$ for $\xi \in L^2(K)$ and $h \in H$ by unimodularity of G , and, noting once again that $\beta_2^G(G)\mathcal{P}_G \subseteq \mathcal{P}_G$, [20, Lemma 2.10] implies

$$\|h \cdot \xi_\alpha - \xi_\alpha\|_{L^2(K)}^2 = \|\beta_2^G(h)\xi_\alpha - \xi_\alpha\|_{L^2(G)}^2 \leq \|\omega_{\beta_2^G(h)\xi_\alpha} - \omega_{\xi_\alpha}\|_{A(G)} = \|h \triangleleft \omega_{\xi_\alpha} - \omega_{\xi_\alpha}\|_{A(G)} \rightarrow 0$$

for all $h \in H$. Let $\xi_\alpha = \xi_\alpha^0 + c_\alpha 1_K$ correspond to the decomposition $L^2(K) = L_0^2(K) \oplus \mathbb{C}1_K$, so that $1 = \|\xi_\alpha^0\|_{L_0^2(K)}^2 + |c_\alpha|^2$ and $\|h \cdot \xi_\alpha^0 - \xi_\alpha^0\|_{L_0^2(K)} \rightarrow 0$ for all $h \in H$. Fix a neighbourhood U of the identity in K with $|K \setminus U| > 0$. If it were the case that $\|\xi_\alpha^0\|_{L_0^2(K)} \rightarrow 0$, then, for α large enough that $|c_\alpha|^2 > \frac{1}{2}$ and $\text{supp}(\omega_{\xi_\alpha}) \subseteq U$, we have for $k \in K \setminus U$ that

$$0 = \omega_{\xi_\alpha}(k) = \omega_{\xi_\alpha^0}(k) + \omega_{\xi_\alpha^0, c_\alpha 1_K}(k) + \omega_{c_\alpha 1_K, \xi_\alpha^0}(k) + |c_\alpha|^2 = \omega_{\xi_\alpha^0}(k) + |c_\alpha|^2$$

because $\xi_\alpha^0 \in L_0^2(K)$, whence

$$-\frac{1}{2}|K \setminus U| > \int_{K \setminus U} \omega_{\xi_\alpha^0}(k) dk = \langle \omega_{\xi_\alpha^0}, \lambda_K(1_{K \setminus U}) \rangle_{A(K), VN(K)} \rightarrow 0,$$

a contradiction. Therefore, passing to a subnet if necessary, we may assume $\|\xi_\alpha^0\|_{L_0^2(K)}$ is bounded away from zero, in which case the vectors ξ_α^0 may be normalized while retaining the property that $\|h \cdot \xi_\alpha^0 - \xi_\alpha^0\|_{L_0^2(K)} \rightarrow 0$ for all $h \in H$. Thus π_K weakly contains the trivial representation. $\square$

The Result

Let K_0 be a nontrivial compact group with normalized Haar probability measure, and let Γ be a discrete group. Put

$$K = K_0^\Gamma$$

with product Haar measure, and let Γ act on K by the Bernoulli shift:

$$(h \cdot x)_g = x_{h^{-1}g} \quad (h, g \in \Gamma, \quad x \in K).$$

Form the locally compact semidirect product $G = K \rtimes \Gamma$.

Theorem 1. *For $G = K_0^\Gamma \rtimes \Gamma$ as above, $A(G)$ is relatively 1-biflat if and only if Γ is amenable.*

Proof Intuition

Crann–Tanko show that relative 1-biflatness forces almost invariant vectors in the Koopman representation of the acting group on $L_0^2(K)$. For a Bernoulli compact shift, $L_0^2(K)$ has a rigid finite-coordinate decomposition. Each finite-coordinate layer is induced from the stabilizer of a finite nonempty subset of Γ , and that stabilizer is finite. Hence the whole Koopman representation is weakly contained in the left regular representation. If it also weakly contains the trivial representation, then the trivial representation is weakly contained in the left regular representation, which is exactly amenability of Γ .

Proof

We first record the Koopman-representation fact used in the theorem.

Lemma 1. *Let $K = K_0^\Gamma$ be a nontrivial compact Bernoulli shift over a discrete group Γ . Let π_K be the unitary representation of Γ on $L_0^2(K)$ induced by the shift action. Then*

$$\pi_K \prec \lambda_\Gamma,$$

where λ_Γ is the left regular representation and $\prec$ denotes weak containment.

Proof. Write

$$L^2(K_0) = \mathbb{C}1 \oplus H_0$$

with $H_0 = L_0^2(K_0)$. Since K_0 is nontrivial, $H_0 \neq 0$. The finite-coordinate cylinder functions give the Hilbert direct sum decomposition

$$L_0^2(K_0^\Gamma) = \bigoplus_{\emptyset \neq F \subseteq \Gamma, F \text{ finite}} H_F, \quad H_F = \bigotimes_{g \in F} H_0,$$

where all coordinates outside F are constants. The Bernoulli shift sends H_F onto H_{hF} .

Fix a finite nonempty subset $F \subseteq \Gamma$, and let $\Gamma_F = \{h \in \Gamma : hF = F\}$. The left translation action of Γ_F on F is faithful: if $hf = f$ for one $f \in F$, then $h = e$. Thus Γ_F embeds into the finite permutation group of F , and is finite. The summand generated by the orbit of H_F is unitarily equivalent to an induced representation

$$\text{Ind}_{\Gamma_F}^\Gamma(\sigma_F),$$

where σ_F is the natural unitary representation of the finite group Γ_F on H_F .

Because Γ_F is finite, it is amenable, and every unitary representation of Γ_F is weakly contained in an amplification of its regular representation. Induction preserves weak containment, and

$$\text{Ind}_{\Gamma_F}^\Gamma(\lambda_{\Gamma_F}) \cong \lambda_\Gamma.$$

Therefore each orbit summand is weakly contained in λ_Γ . Taking the Hilbert direct sum over all finite nonempty F -orbits gives $\pi_K \prec \lambda_\Gamma$. $\square$

We now prove the theorem. If Γ is amenable, then $K_0^\Gamma \rtimes \Gamma$ is compact-by-amenable, hence amenable. By Losert–Rindler, amenable locally compact groups are QSIN, and by the standard QSIN sufficiency result used in Crann–Tanko, $A(G)$ is relatively 1-biflat.

Conversely, suppose $A(K \rtimes \Gamma)$ is relatively 1-biflat. If Γ were nonamenable, then Γ would be infinite, so $K = K_0^\Gamma$ would be an infinite compact group. Crann–Tanko Proposition 4.3 applies to the semidirect product $K \rtimes \Gamma$ and implies that the Koopman representation π_K on $L_0^2(K)$ weakly contains the trivial representation 1_Γ . By the lemma, $\pi_K \prec \lambda_\Gamma$. Weak containment is transitive, so

$$1_\Gamma \prec \lambda_\Gamma.$$

By Hulanicki’s characterization, this is equivalent to amenability of Γ . This contradiction proves that Γ must be amenable.

Limitations and Novelty Check

This packet does not solve the full Crann–Tanko conjecture. It proves the conjecture for compact Bernoulli wreath products, including the standard non-QSIN cases $K_0^\Gamma \rtimes \Gamma$ with Γ nonamenable.

Searches performed on 2026-07-03 included the phrases “relative 1-biflat Fourier algebra QSIN”, “Bernoulli relative 1-biflat Fourier algebra”, “wreath product relative 1-biflat Fourier algebra”, “ K^Γ relatively 1-biflat”, and “Koopman relative biflatness Fourier algebra”. These searches found the source paper but did not surface this Bernoulli-wreath characterization.

Human Review Recommendation

Review the finite-coordinate decomposition and the weak-containment step for arbitrary compact K_0 , especially if nonmetrizable compact groups or uncountable Γ are allowed. The proof is standard Hilbert-space representation theory, but that is the natural place for a human reviewer to check set-theoretic conventions.

References

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- [2] O. Yu. Aristov, V. Runde, and N. Spronk, *Operator biflatness of the Fourier algebra and approximate indicators for subgroups*, J. Funct. Anal. 209 (2004), 367–387.
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